

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

Section - 1

1. A car covers the first half of the total distance of its journey with a velocity v_1 . The remaining half of the distance is covered with a velocity v_2 for half of the remaining time and with a velocity v_3 for the other half of the remaining time. What is the average velocity of the car for the entire journey?

A) $\frac{v_1 + v_2 + v_3}{3}$

B) $\frac{2v_1(v_2 + v_3)}{2v_1 + v_2 + v_3}$

C) $\frac{v_1(v_2 + v_3)}{v_1 + v_2 + v_3}$

D) $\frac{3v_1v_2v_3}{v_1v_2 + v_2v_3 + v_3v_1}$

2. During a laboratory experiment, a student notices that while boiling water, the thermometer reading remains stuck at 100°C even though the burner is still on at maximum heat. What is the most scientifically accurate reasoning for this?

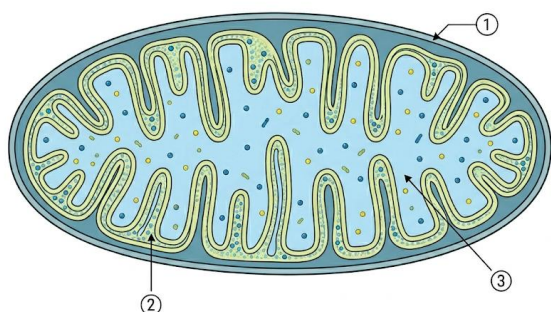
A) The thermometer has reached its maximum physical capacity.

B) The heat energy is being used to overcome the intermolecular forces of attraction rather than increasing kinetic energy.

- C) The surrounding air is cooling the water at the exact same rate the burner is heating it.
- D) Water molecules are reflecting the heat energy back into the container.
3. During an experiment, a researcher treated a plant cell with a specific chemical that inhibits the function of the Golgi apparatus. Which of the following cellular processes would be most immediately and directly affected in this plant cell?
- A) The synthesis of ATP during aerobic respiration in the mitochondria.
- B) The primary transcription of RNA from the DNA template in the nucleus.
- C) The formation of the cell plate during cytokinesis and the packaging of glycoproteins.
- D) The movement of water through the aquaporins located on the plasma membrane
4. The work done by a variable force is best represented by:
- A) Slope of F - x graph
- B) Area under F - x graph
- C) Area under v - t graph
- D) Slope of x - t graph
5. A mixture contains: iron + sand + salt + kerosene
- By Following the flowchart how could you separate the mixture.
- Step1: Magnet
- Step2: Addwater + filtration
- Step3: Separating funnel
- Step 4: Evaporation
- What is removed in Step 3?

- A) Sand
- B) Salt
- C) Kerosene
- D) Iron
6. A scientist observes a cell under an electron microscope and notices an unusually high density of mitochondria and a very large, prominent nucleolus. Based on these characteristics, what can be inferred about the cell's primary function?
- A) The cell is highly active in protein synthesis and requires a large amount of energy for metabolic processes.
- B) The cell is likely a dormant storage cell, such as a starch-storing parenchyma cell in a root.
- C) The cell is a mature human red blood cell (RBC) designed for maximum oxygen transport.
- D) The cell is a protective skin cell that has ceased all internal metabolic activity.
- A vehicle moving with a speed v on a straight road can be stopped by applying brakes within a minimum distance s . If the vehicle's speed is increased to $3v$ and the exact same maximum retarding force is applied by the brakes, what will be the new minimum stopping distance?
- A) $6s$
- B) $9s$
- C) 3
- D) s
7. An element X has an atomic number of 15. It reacts with element Y, which requires two electrons to achieve a stable octet. What is the most likely formula of the compound formed between X and Y, and what does this tell you about their valencies?
- A) XY_2 ; Valency of X is 2, Y is 1.
- B) X_2Y_3 ; Valency of X is 3, Y is 2.
- C) X_3Y_2 ; Valency of X is 2, Y is 3.
- D) XY_5 ; Valency of X is 5, Y is 1.

8. Consider a diagram of a Mitochondrion. Why is structure (2) folded rather than being a simple smooth sphere?



- A) To provide more space for the storage of hereditary DNA.
 B) To decrease the volume of the organelle so more can fit in the cell.
 C) To increase the surface area for chemical reactions that generate ATP.
 D) To trap sunlight more efficiently for the process of glycolysis.
9. If the Earth is considered a homogeneous sphere with mean density ρ , radius R , and mass M , what is the correct formula for the acceleration due to gravity (g) at the surface of the Earth?
- A) $g = \frac{4}{3}\pi\rho G(R - d)$ B) $g = \frac{4}{3}\pi\rho GR$
 C) $g = \sqrt{2GR}$ D) $g = \frac{GMm}{R^2}$
10. Look at the table below for four particles. Which particle represents a negative ion (anion)?

Particle	Protons	Neutrons	Electrons
P	11	12	11
Q	17	18	18
R	12	12	10
S	1	0	0

- A) Particle P B) Particle Q
 C) Particle R D) Particle S

11. While observing a slide of a vascular bundle, a student notices a specific complex tissue. It consists of long, tube-like structures with perforated end walls called sieve plates. Nearby, there are specialized cells with dense cytoplasm and large nuclei that seem to control the activity of these tubes. If these specialized "helper" cells were damaged, what would be the primary consequence?

- A) The upward movement of water and dissolved minerals from the roots would cease entirely.
 B) The plant would be unable to provide mechanical support, causing the stem to collapse.
 C) The plant would lose its ability to undergo secondary growth and increase in girth.
 D) The translocation of sucrose and other organic solutes would be severely disrupted.

12. A section of a mammalian organ reveals an epithelial lining where the cells are as tall as they are wide, appearing square-like in a cross-section. This tissue is found in the lining of kidney tubules and salivary gland ducts. What is the primary functional advantage of this specific shape?

- A) It provides mechanical strength and is specialized for secretion and absorption.
 B) It provides a very thin surface to allow for the rapid diffusion of gases like oxygen.
 C) It allows the tissue to stretch and expand significantly without tearing.
 D) It uses hair-like projections to move mucus across the surface of the organ.

13. A poultry farmer wants to increase the production of "Broilers" for meat purposes. He is advised to cross-breed a local Indian variety (like Aseel) with an exotic variety (like Leghorn). What are the specific traits he should be looking for in the resulting hybrid to ensure a high-profit margin?

- A) High number of eggs produced per year and large body size.
- B) Low fiber requirement in diet and high tolerance to high temperatures.
- C) Ability to produce high-quality silk and resistance to viral diseases.
- D) Fast growth rate, low mortality, and high efficiency in converting feed into meat.

14. A farmer decides to switch from using chemical fertilizers (NPK) to using "Green Manure." He grows a crop of Sunn hemp and mulches it back into the soil before sowing his main Rice crop. Compared to chemical fertilizers, what is a distinct disadvantage of using only green manure in the short term?

- A) It reduces the organic matter (humus) content of the soil, making it sandy.
- B) It releases nutrients slowly and in lower concentrations compared to the immediate "hit" of chemical fertilizers.
- C) It causes "Eutrophication" in nearby water bodies much faster than urea.
- D) It makes the soil toxic to earthworms and other beneficial microorganisms.

Section - 2

15. An astronaut of mass 60 kg is inside a satellite orbiting Earth at radius $r = 2R$ (R = Earth's radius). Which of the following statements are correct?

- A) Gravitational force on astronaut is zero
- B) Apparent weight of astronaut is zero
- C) Acceleration of astronaut is $\frac{GM}{(2R)^2}$
- D) Normal reaction acting on astronaut is zero

16. The Aufbau Principle suggests that electrons fill orbitals of lower energy first. In which scenarios might the observed configuration appear to deviate from the standard rule?

- A) When an atom is in an excited state due to absorption of a photon.
- B) In a perfectly isolated hydrogen atom where energy depends only on n .
- C) Whenever an atom forms a covalent bond with another atom.
- D) In transition metals where the d and s orbitals are very close in energy.

17. A plant cell is placed in a solution that is significantly more concentrated (hypertonic) than its own cell sap. Which of the following physiological changes will occur over time?

- A) The cell wall will shrink inward until it touches the nucleus.
- B) The protoplast will lose water and shrink away from the cell wall.
- C) The vacuole will decrease in size as water moves out via osmosis.
- D) The cell will eventually burst due to the high external osmotic pressure.

18. A person pushes a box inside a moving train with constant force F .

- A) Work done in train frame is $F \cdot s$
- B) Work done in ground frame is $F \cdot (s + s_0)$
- C) Work done is same in all frames
- D) Work depends on displacement, which can vary with frame

19. If you have 1 mole of Ethane (C_2H_6) and 1 mole of Water (H_2O), what do they have in common?

(Atomic masses: $C=12$, $H=1$, $O=16$)

- A) They both contain the same number of molecules.
- B) They both have the same total mass in grams.
- C) They both represent one "gram molecule" of their respective substances.
- D) They both contain the same total number of atoms

20. Considering the structure and function of the Plasma Membrane, which of the following statements accurately describe its "Fluid Mosaic" nature?

- A) The "mosaic" part refers to the various proteins embedded in or attached to the phospholipid bilayer.
- B) The membrane is a static, solid structure that prevents any movement of lipids.
- C) The "fluid" part refers to the ability of lipids and proteins to move laterally within the layer.
- D) Carbohydrates are attached to the inner surface of the membrane to provide energy.

21. Epithelial tissues form the "first line of defense" in the animal body. Which of the following pairs correctly match the epithelial type with its specific location and function?

A) Squamous Epithelium — Lining of blood vessels — Facilitates transport through a thin, flat surface.

B) Ciliated Columnar Epithelium — Respiratory tract — Moves mucus forward to clear the passage.

C) Stratified Squamous Epithelium — Skin — Arranged in many layers to prevent wear and tear.

D) Cuboidal Epithelium — Lining of the small intestine — Primary site for the exchange of gases.

22. A scientist is evaluating the "Environmental Impact" of using heavy doses of chemical fertilizers and pesticides over ten years. Which of the following phenomena are likely consequences of this intensive farming?

A) Eutrophication of nearby ponds leading to the depletion of dissolved oxygen and death of aquatic life.

B) Biological Magnification, where toxic pesticide residues increase in concentration at each higher trophic level.

C) Increased soil fertility and a permanent rise in the population of nitrogen-fixing bacteria.

D) Destruction of soil structure and a decrease in the organic matter (humus) content over time.

Section - 3

COMPREHENSION-1

Forces dictate how objects move, deform, and interact, whether through direct physical contact like springs or at a distance like gravity. By understanding concepts like momentum and inertia, we can accurately predict how bodies will

behave when these forces are applied or removed.

23. Why is a passenger thrown outwards when a car takes a sudden, sharp curve?

- A) Unbalanced contact forces pull them.
- B) Inertia of motion accelerates them forward.
- C) Inertia of direction makes their body try to keep moving in a straight line.
- D) The normal reaction of the seat pushes them away.

24. In the spring force equation $F = -Kx$, what does the x explicitly represent?

- A) The spring constant
- B) The change in the spring's length
- C) The mass of the spring
- D) The time taken to stretch it

25. If a heavy truck and a tiny bullet have the exact same momentum, which must be logically true?

- A) The bullet must have a much higher velocity to compensate for its small mass.
- B) The truck must be moving much faster than the bullet.
- C) Both will apply zero impulse upon impact.
- D) The truck will have less inertia of rest.

COMPREHENSION-2

The Zero-Waste Textile Factory

The engineering team employs a multi-stage purification process:

Stage A: The water is held in large tanks where heavy sand particles settle at the bottom over time.

Stage B: The top layer of oil, which does not mix with water, is skimmed off using a specialized mechanical barrier.

Stage C: The remaining colored water is passed through a sophisticated membrane that allows water molecules to pass but traps the larger dye molecules.



26. In Stage A of the process described above, the factory utilizes the principle of sedimentation. Which of the following would speed up this specific stage if the particles were too fine to settle quickly?

- A) Increasing the temperature of the water.
- B) Adding alum to initiate flocculation.
- C) Stirring the mixture vigorously.
- D) Passing the mixture through a magnetic separator

27. A technician suggests that Stage B could be replaced by using a separating funnel. Why is this reasoning technically correct but practically difficult for the factory?

- A) Separating funnels only work for solids dissolved in liquids.
- B) A separating funnel requires the liquids to have the exact same density.
- C) Separating funnels are designed for small-scale laboratory use, not continuous industrial flow.
- D) Oil and water cannot be separated by a separating funnel because they form a homogenous mixture.

28. If the "dissolved dyes" in Stage C were actually heat-stable and had a much higher boiling point than water, which alternative technique could recover both the pure water and the solid dye?

- A) Simple Distillation
- B) Fractional Distillation
- C) Centrifugation
- D) Winnowing

COMPREHENSION-3

A delivery drone first moves straight for In a rural developmental project, a cooperative dairy farm is trying to bridge the gap between "Yield" and "Resilience." They have two groups of cattle:

- **Group A:** Exotic breeds like **Holstein-Friesian**, which produce an average of 25 liters of milk per day but are highly susceptible to "Tick Fever" and heat stress.
- **Group B:** Local Indian breeds like **Sahiwal**, which produce only 8 liters per day but show remarkable resistance to tropical diseases and can thrive in temperatures exceeding 40°C.

The farm decides to implement a "Cross-breeding" program. After several years, they produce a hybrid generation. To maintain the health of this new herd, they provide a diet consisting of 70% "Roughage" (hay and silage) and 30% "Concentrates" (oil cakes, cereals, and minerals).

29. What is the biological objective of creating the hybrid between Group A and Group B?

- A) To create a new species of cattle that can survive without any water intake.

B) To combine the long lactation period and high yield of the exotic breed with the disease resistance of the local breed.

C) To ensure that the cattle produce milk that is naturally fortified with plant-based proteins.

D) To reduce the "Roughage" requirement of the cattle to zero to save on fodder costs.

30. The farm manager emphasizes the use of "Concentrates" in the hybrid's diet. What is the specific functional role of these concentrates?

A) They provide the bulk required to keep the stomach full and prevent hunger.

B) They are primarily used to increase the water-holding capacity of the cow's tissues.

C) They provide high levels of proteins and nutrients required for high milk production.

D) They act as a natural antibiotic to kill the parasites that cause Tick Fever.

31. The main objective of the dairy farm's cross-breeding program is to:

A) Increase only milk yield

B) Improve only disease resistance

C) Combine high yield with better resilience

D) Reduce the number of cattle

Section - 4

32.

COLUMN A	COLUMN B
A) $W = \int F dx$	P) Zero for $\theta = 90^\circ$
B) $W = F \cos \theta$	Q) Work equals area under F-x graph
C) $W_{A \rightarrow B} + W_{B \rightarrow A} = 0$	R) Conservative force property

D) $W = F \cdot s$ (frame dependent displacement)	S) Changes with frame of reference
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- A) A-Q, b-P, c-R, d-S
 B) A-P, b-Q, c-S, d-R
 C) A-Q, b-R, c-P, d-S
 D) A-R, b-P, c-Q, d-S

33.

COLUMN -A	COLUMN-B
A) Sublimation of Dry Ice (CO_2)	P) Intermolecular space is minimum; particles only vibrate about a mean position.
B) Water vapor condensing on a cold glass	Q) Molecules overcome atmospheric pressure to move directly from a fixed lattice to a random motion.
C) Mercury in a thermometer at room temperature	R) Kinetic energy decreases; molecules lose speed and move closer together to form a definite volume.
D) A block of Lead at room temperature	S) Particles have enough energy to slide over each other but remain attracted to the bulk..

- A) A-R, B-P, C-Q, D-S
 B) A-P, B-R, C-S, D-Q
 C) A-R, B-P, C-S, D-Q
 D) A-Q, B-R, C-S, D-P

34.

COLUMN -A	COLUMN-B
A) Smooth ER	P) Autolysis and cellular digestion
B) Golgi Apparatus	Q) Lipid synthesis and detoxification
C) Lysosomes	R) ATP generation via oxidation
D) Mitochondria	S) Modification and packaging of proteins

- A) A - P, B - Q, C - R, D - S
 B) A - Q, B - P, C - R, D - S
 C) A - R, B - S, C - Q, D - P
 D) A - Q, B - S, C - P, D - R